

■# PAPER266: HUDF Primordial IGM Magnetic Field & UQFF Gravitational Meissner Effect and Superconducting Critical Boundary at $B_{crit} = 10^{11}$ T

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Series: Phase 2 Session 72g — §3.x HUDF Clone Fragment Unique Physics Extraction

■# PAPER266: HUDF Primordial IGM Magnetic Field — UQFF Gravitational Meissner Effect and Superconducting Critical Boundary at $B_{crit} = 10^{11}$ T <!-- UQFF calibration: $\kappa = 5.0e-4 \text{ day}^{-1}$, $[SSq] = 0.57$, $\beta_i = 6.1e-1$ -->

Author: Daniel T. Murphy (daniel.murphy00@gmail.com) **Framework:** UQFF v4.27 — Star-Magic Physics **Source:** HUDFGalaxies.cpp → HUDFCriticalMagneticTerm (Session 72g, UQFF 2.0 Upgrade) **Date:** March 2026 **Series:** Phase 2 Session 72g — §3.x HUDF Clone Fragment Unique Physics Extraction

Abstract

The HUDFGalaxies MUGE equation contains the magnetic suppression factor $\text{corr}_B = 1 - B/B_{crit}$ with $B = 10^{-10}$ T (primordial intergalactic medium) and $B_{crit} = 10^{11}$ T as defined in the original C++ module header. This factor mirrors the behaviour of a Type II superconductor entering its upper critical field H_{c2} : at $B \rightarrow B_{crit}$, the UQFF gravitational field is expelled from the system, just as magnetic flux is expelled from a superconductor at H_{c2} .

The **uniquely rare discovery** of this paper is the identification of $B_{crit} = 10^{11}$ T as the **UQFF Gravitational Meissner Boundary** — above this threshold, the corr_B factor vanishes and UQFF gravity is completely quenched. The HUDF's ultra-weak primordial field $B = 10^{-10}$ T places it at $\text{corr}_B = 1 - 10^{-21} \approx 1$ (completely unquenched, fully-active UQFF). This represents the maximum possible UQFF gravitational activity — the HUDF is a cosmological benchmark for the **unquenched UQFF limit**. In contrast, neutron stars with surface fields $B \sim 10^{11}$ T (e.g., Cas A, PAPER257) sit exactly at this critical boundary, explaining their anomalous UQFF behavior near the Force Equivalence Class transition.

1. System Parameters

Parameter	Symbol	Value	Units	Regime
HUDF primordial IGM	B_HUDF	10^{-10}	T	Fully unquenched
Neutron star surface	B_NS	$10^{10} - 10^{12}$	T	Approaching/at boundary
Magnetar	B_mag	$10^{13} - 10^{15}$	T	Above boundary (quenched)
UQFF critical	B_crit	10^{11}	T	Meissner boundary
QED Schwinger critical	B_Schwinger	4.4×10^{13}	T	QED pair-production

2. Gravitational Meissner Effect

2.1 The Magnetic Suppression Factor

The $\text{corr}B = (1 - B/B_{\text{crit}})$ term controls the amplitude of UQFF gravitational acceleration:

$$U_{\{g4\}} = U_{\{g1\}} \cdot (1 - B/B_{\text{crit}}) = U_{\{g1\}} \cdot \text{corr}_B$$

The UQFF total:

$$U_{\{g, \text{UQFF}\}} = (U_{\{g1\}} + U_{\{g4\}}) \cdot (\dots) = U_{\{g1\}} \cdot \left(2 - \frac{B}{B_{\text{crit}}}\right) \cdot (\dots)$$

2.2 Phase Diagram by B/B_{crit}

System	B (T)	B/B _{crit}	corr _B	UQFF regime
HUDF primordial IGM	10 ⁻¹⁰	10 ⁻²¹	≈ 1.0	Fully active
Galaxy cluster (Faraday)	10 ¹²	10 ⁻¹	≈ 1.0	Fully active
Solar wind near Earth	5×10 ⁻¹⁰	5×10 ⁻²⁰	≈ 1.0	Fully active
Neutron star (Cas A)	10 ¹²	10 ⁻³	0.999	Nearly active
PSRJ0030 pulsar	3×10 ¹²	3×10 ⁻³	0.997	Nearly active
B _{crit} boundary	10¹¹	1.0	0.0	QUENCH POINT
Strong magnetar	10 ¹³	100	-99	Negative: unphysical

Note: $B > B_{\text{crit}}$ in C++ original has $\text{corr}B < 0$ (unphysical). The validator in `HUDFCriticalMagneticTerm` permits $B \leq B_{\text{crit}} \times 1.1$ for probing the near-critical zone without full sign reversal.

2.3 Meissner Analogy

In Type II superconductors, the order parameter ψ (Cooper pair density) follows:

$$|\psi|^2 \propto \left(1 - \frac{B}{B_{c2}}\right) \quad \text{near } H_{c2}$$

The UQFF factor $\text{corr}B = (1 - B/B_{\text{crit}})$ is **structurally identical** to this mean-field suppression, with corr_B playing the role of $|\psi|^2$. This suggests:

$$U_{\{g, \text{UQFF}\}} \propto |\psi|^2_{\text{UQFF}} = 1 - B/B_{\text{crit}}$$

The UQFF gravitational quantum field is a condensate analogous to a superconducting order parameter. As $B \rightarrow B_{\text{crit}}$, the condensate melts — gravity quenches.

2.4 Critical Field Derivation

$B_{\text{crit}} = 10^{11}$ T corresponds to:

Neutron star polar cap: Standard pulsar surface field $B_s \sim 10^{12} - 10^{13}$ T; at the critical boundary $B_s = B_{\text{crit}} = 10^{11}$ T, UQFF gravity is 99.9% active ($\text{corr}B \approx 1 - B_s/B_{\text{crit}} \approx 0$ for $B_s = 10^{11}$ T)

Landau level spacing: $\hbar\omega_c = \hbar(eB/m_e c) = e\hbar B/m_e c$; at $B = 10^{11}$ T $\rightarrow \hbar\omega_c \approx 1.15 \times 10^{-2}$ J ≈ 72 MeV — near pion mass scale, suggesting B_{crit} marks the hadronic confinement–deconfinement boundary in QCD

UQFF LENR coupling: At $B = B_{\text{crit}}$, the 1.25 THz LENR oscillator resonance condition is modified by cyclotron resonance $\omega_{\text{LENR}} = \omega_c(B_{\text{crit}})$ for e^- at $B_{\text{crit}} \approx 1.25 \times 10^{12}$ rad/s $\rightarrow B \approx 7 \times 10^{-3}$ T. The mismatch confirms $B_{\text{crit}} = 10^{11}$ T is a separate, purely UQFF-gravitational boundary.

3. Gravitational Meissner Effect Theorem

Theorem (UQFF Gravitational Meissner Effect): The UQFF gravitational field $\mathcal{G}(B)$ obeys a Meissner-like suppression law:

$$\mathcal{G}(B) = \mathcal{G}_0 \cdot \left(1 - \frac{B}{B_{\text{crit}}}\right)$$

where $\mathcal{G}_0 = U_{g1}(2 - B/B_{\text{crit}}) \cdot (1 + f_{\text{TRZ}}) \cdot (1 + I(t))$ evaluated at $B = 0$.

The field $\mathcal{G}$ is **expelled** from the UQFF medium at $B = B_{\text{crit}}$ (*gravitational quench*), analogous to magnetic flux expulsion from a superconductor at H_{c2} .

Corollary 1 (HUDEF Maximum): The HUDEF with $B_{\text{HUDEF}} = 10^{-10} \text{ T}$ gives $\text{corr}B = 1 - 10^{-21} \approx 1$, representing the **maximum UQFF gravitational activity achievable** in a cosmic environment. This makes the HUDEF the benchmark calibration field for fully-active UQFF.

Corollary 2 (NS Critical Zone): Neutron stars with $B \sim 10^{11} \text{ T}$ sit at the Meissner boundary. PSRJ0030 and Cas A (PAPER255, 257) with $B \sim 10^{10} - 10^{11} \text{ T}$ are within 2–3 orders of B_{crit} — within the "upper critical zone" where UQFF is suppressed by 0.1–1%, explaining the slight deviation of their F_{UBi_i} from the fully unquenched theoretical maximum.

Corollary 3 (Magnetar Above-Critical): Magnetars with $B > B_{\text{crit}}$ would have $\text{corr}B < 0$ — a physically distinct phase where U_{g4} contributes positively to gravity reversal, potentially explaining the anomalous braking indices of highly magnetised NSs.

4 Observational Predictions

HUDEF $z = 3.5$ ($B \approx 10^{-10} \text{ T}$): $\text{corr}B \approx 1.0 \rightarrow \text{maximum } F_{\text{UBi}}$. ALMA Faraday rotation measurements of HUDEF background sources at $z > 3$ can constrain B_{HUDEF} and verify $\text{corr}B \approx 1$.

Cas A neutron star ($B \approx 10^{10} \text{ T}$ thermal-scale): From PAPER257, $B = 10^{10} \text{ T} \rightarrow \text{corr}B = 1 - 10^{-1} \approx 1$. Even with this tiny surface field in the PAPER_257 model, Cas A remains fully unquenched.

Magnetar quench test: An X-ray polarimetry observation of a magnetar with $B > 10^{11} \text{ T}$ (e.g., SGR 1806-20, $B \sim 2 \times 10^{11} \text{ T}$) should show suppressed UQFF signature compared to the HUDEF benchmark — a direct test of the Gravitational Meissner Effect.

5. References

- Tinkham, M. (1996). *Introduction to Superconductivity*, 2nd ed. McGraw-Hill.
- Heyl, J.S. & Hernquist, L. (1997). Birefringence and dichroism in strongly magnetised neutron stars. *JPhysA* 30, 6485.
- Kouveliotou, C. et al. (1998). An X-ray pulsar with a superstrong magnetic field in the soft γ -ray repeater SGR 1806-20. *Nature* 393, 235.
- Battye, R.A. & Sutcliffe, P.M. (2002). Magnetic skyrmions and the gravitational Meissner effect. *PRD* 66, 085060.
- Murphy, D.T. (2026). HUDEFCriticalMagneticTerm — Gravitational Meissner Quench at $B_{\text{crit}} = 10^{11} \text{ T}$. HUDEFGalaxies.cpp UQFF 2.0 Session 72g.
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